How-To Library

Informatica

Install Enterprise Data Preparation on Docker with the Container Utility (10.4.1)

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Abstract

Informatica provides the Informatica container utility to install the Informatica domain quickly. This article describes how to install Enterprise Data Preparation from the Docker image through the Informatica container utility on Docker.

Supported Versions

- Enterprise Data Preparation 10.4.1

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Overview

With a web-based interface in the Informatica container utility, you can easily create standard docker images and create a container using docker.

Docker is an open source platform that provides an isolated environment called containers to run the applications. Docker allows independent containers to run within a single Linux instance. A docker image is an executable package that can run an application, a code, run-time files, environment variables, or configuration files. A container is a run-time instance of an image.

Run the Informatica container utility to build or run the docker image to configure the domain. You can build the docker image with the base operating system and Informatica product binaries. With the Informatica container utility, you can create containers using this docker image to configure the Informatica domain.

When you run the docker image, you can install Informatica products with the Informatica container utility and you can create or join a domain.

With the Informatica container utility, you can build or run the docker image with the silent installer. CI/CD or continuous integration and continuous delivery, is a practice that automates the integration and delivery operations in a CI/CD pipeline. With the CI/CD pipeline, you can quickly deploy, test, and deliver the docker image to the production environment. You can generate the docker image from the CI/CD contents. The CI/CD content uses the same input values that you used in the Informatica container utility to either build or run the docker image. You can manually replace the sample silent input properties with the CI/CD contents before you run the silent installation.

You can create the application services when you run the docker image, or you can create them in the Administrator tool.

Container Utility Process Checklist

Perform the following tasks associated with the installation:

- ❑ Plan for all installation components within the domain, such as nodes and services.
- ❑ Complete prerequisites:
  - Verify system requirements.
  - Configure docker engine.
  - Set up keystore and truststore files.
  - Extract the Informatica container utility and access the Informatica installer .tar file.

- Verify the license key.
- Set up repository databases.
- Generate CI/CD.

❑ Build the Informatica docker image with base operating system and Informatica binaries.

❑ Run the docker image to configure the Informatica domain.

❑ Deploy the docker image quickly. Or, download and run the default image.

❑ Build or run the docker image in silent mode with CI/CD.

❑ Complete the post-requisites:

- Complete the domain configuration.
- Create the application services.

Before You Begin

Before you install the Informatica container utility, verify that the machine meets the pre-installation requirements for the Informatica product installation.

Verify System Requirements

Verify that your environment meets the minimum system requirements for the installation process:

- Disk space of 50 GB.
- Free disk space of 100 GB for the working directory when you build the image.
- Supported image name or one of the following image names for the subscribed base operating system: `registry.access.redhat.com/rhel7` or `centos:7`.
- Register and subscribe on page 26 the machine where you build the docker images.
- Temporary disk space and permissions.
- Port requirements
- Sizing Requirements

Verify System Requirements

Verify that your environment meets the minimum system requirements for the installation process:

- Supported image name or one of the following image names for the subscribed base operating system: r`egistry.access.redhat.com/rhel7 or centos:centos7`.
- Register and subscribe the machine where you build the docker images.
- If IPv4 forwarding is not enabled on your system, add the `net.ipv4.ip_forward=1` parameter in the following file: `/etc/sysctl.conf`.
  To restart the network service, run the following command:

  ```
  systemctl restart network
  ```

  To verify the system configuration, run the following command:

  ```
  sudo sysctl -p
  ```

Set Up the Keystore and Truststore Files

When you install the Informatica services, you can configure secure communication for the domain and set up a secure connection to Informatica Administrator (the Administrator tool). If you configure the security options, you must set up the keystore and truststore files.

If you plan to enable Kerberos or SSL authentication for the cluster, the keystore and truststore files must be located in the machine that hosts the Informatica domain.

Contact Informatica Global Customer Support

When you initially contact Informatica Global Customer Support for Informatica installation access, the Informatica Global Customer Support team initiates a shipping request. The shipping team sends the Informatica installation link and license information to you.

You can contact Informatica Global Customer Support online for access to the following files:

- Informatica installer .tar file
- License key
- Informatica client installation

Access to .tar File

You require access to both the Informatica container utility and the Informatica installer .tar file to install the product.

Download and extract the Informatica container utility installer files from the download center for Enterprise Data Preparation location on Informatica Marketplace.

After you extract the informatica_container_utility_linux-x64.zip file, contact the support team to access the Informatica installer .tar file. You can then place the Informatica installer tar file on the machine where you build the image.

To gather inputs for the silent input properties files, launch the Informatica container utility and click **Generate silent properties for CI/CD**. You can then change the format of the properties file to UNIX before you run the silent installation.

Verify the License Key

Before you install the software, verify that you have the license key available for the Informatica product installer.

The product is provided as a Bring Your Own License (BYOL). Copy the license key file to a directory accessible to the user account that installs the product. You can use an existing Informatica license or contact Informatica Global Customer Support if you do not have a license key or if you have an incremental license key.

The volume directory is the directory where the Informatica license key and certificates are stored. You must have read, write, and execute permissions on the volume directory.

Note: You do not need a license for the Informatica container utility.

Access the Informatica Client Installation

Ensure that the Informatica Global Customer Support team has provided you with access to the Informatica client installation.

Before you install the Informatica clients, verify that the machine meets the minimum system and third-party software requirements. If the machine where you install the Informatica clients is not configured correctly, the installation can fail.

Set Up Databases

Informatica components store metadata in relational database repositories. When you set up the databases, you will need to allow for disk space and create the databases with parameters required by the product. You will also create user accounts and install the database clients.

Set up databases for the following repositories:

Domain configuration repository

The domain stores configuration and user information in a domain configuration repository. You can create the domain configuration repository on an Oracle, PostgreSQL, or Microsoft SQL Server database.

Model repository

The Model repository stores information about the metadata for the data objects and mappings in a relational database. If you want to generate monitoring statistics, you must create a dedicated monitoring Model repository to store run-time monitoring statistics. You can create the Model repository and the monitoring Model repository on an Oracle, PostgreSQL, or Microsoft SQL Server database.

Create either an on-premises database or a container database for each repository. When you run the docker image, you will provide the database information. Create the container database to share the system resources for disk space, kernels, and operating system libraries within the containerized application.

Domain Configuration Repository

The domain stores configuration and user information in a domain configuration repository.

You can create the domain configuration repository in one of the following databases.

Oracle

Complete the following tasks to prepare the Oracle database on-premises or in the docker container:

- Create the database with the required parameters, allowing for 200 MB of disk space.
- Set up database user accounts.
- Install compatible versions of the Oracle client and Oracle database server. You must also install the same version of the Oracle client on all machines that require it.

PostgreSQL

Complete the following tasks to prepare the PostgreSQL database on-premises or in the docker container:

- Create the database with the required parameters, allowing for 200 MB of disk space.
- Set up database user accounts.

You do not need to install the PostgreSQL client software, as the installer bundles it with the image.

Microsoft SQL Server

Complete the following tasks to prepare the Microsoft SQL Server database on-premises or in the docker container:

- Create the database with the required parameters, allowing for 200 MB of disk space.
- Set up database user accounts.
- Download the Microsoft SQL Server client and install it on all machines that require it.

Model Repository

Create a Model repository to store information about the metadata for the data objects and mappings in a relational database. If you want to generate monitoring statistics, you must create a dedicated monitoring Model repository to store run-time monitoring statistics.

You can create the Model repository and the monitoring Model repository in one of the following databases:

Oracle

Complete the following tasks to prepare an Oracle database for the Model repository and for the monitoring Model repository or to create a container database:

- Create the database with the required parameters, allowing for 200 MB of disk space.
- Set up database user accounts.
- Install compatible versions of the Oracle client and Oracle database server. You must also install the same version of the Oracle client on all machines that require it.

PostgreSQL

Complete the following tasks to prepare a PostgreSQL database for the Model repository and for the monitoring Model repository or to create a container database:

- Create the database with the required parameters, allowing for 200 MB of disk space.
- Set up database user accounts.

You do not need to install the PostgreSQL client software, as the installer bundles it with the image.

Microsoft SQL Server

Complete the following tasks to prepare a Microsoft SQL Server database for the Model repository and for the monitoring Model repository or to create a container database:

- Create the database with the required parameters, allowing for 200 MB of disk space.
- Set up database user accounts.
- Download the Microsoft SQL Server client and install it on all machines that require it.

Generate CI/CD

Before you can run the silent installer to build or run the docker image, click **Generate silent properties for CI/CD** from the Informatica container utility. Use the generated CI/CD values to replace the contents of the silent input properties file. You must convert the properties file to the UNIX format before you run the silent installer.

When you run generate CI/CD, you can copy all the user input values entered to build or run the docker image from the Informatica container utility from a single panel. The generated CI/CD contents are in the supported format of the

silent install property files. Manually save the CI/CD contents into the silent input properties file located in the silent installation directory. Convert the properties file to the UNIX format before you run the silent installation.

Build the Informatica Docker Image

Build a docker image for Informatica products. After you build the docker image, the Informatica container utility uses the docker image to run on containers. Ensure that the Informatica docker image is stored in the local host or docker registry before you install Informatica products in the docker with the Informatica container utility.

Step 1. Choose the Deployment Type

On the **Deployment type** page, you must select the deployment type and host instance to build the docker image.

1. Launch the Informatica container utility with the `sh startup.sh` command. By default, the utility uses port 12386.

   To specify the startup command with a different port number, enter the values in the following format: `sh startup.sh <required port number>`
2. To build the Informatica Docker image, select **Build the docker image**.

   The **Deployment type** page appears.
3. On the **Deployment type** page, enter the information for the deployment type.
   - Select Enterprise Data Preparation for the product.
   - Choose to host on Amazon Web Services (AWS), Microsoft Azure, or on-premises.

   The build image uses Docker as the default deployment type.
4. Optionally, you can choose to provide secure authentication to build the image or deploy the containers on the remote host. If you do not want to set the secure authentication for the containers, go to step 6.
5. If you enable secure authentication, set the authentication type to **Password** or **Key**.
   - For password type authentication, enter the machine IP address or host name, user name, password, and port to connect to the host machine.
   - For key type authentication, enter the machine IP address or host name, user name, path of the host authentication key, and port number.
6. Click **Next**.

   The **Build the docker image** page appears.

Step 2. Build the Docker Image

On the **Build Docker Image** page, enter information for the tar file location and the images, or accept the default values to build the docker image.

1. Specify the base operating system path to build the docker image.
   - To build the docker image on CentOS, enter a value or select the default value of `centos:7`.
   - To build the docker image on RHEL, enter a value or select the default value of `registry.access.redhat.com/rhel7:7.8`.
2. Enter the Informatica installer tar file path on the machine where you build the image.

   The Informatica container utility uses the tar utility to extract the installer server files to a directory on your machine where you build the image.
3. Specify the docker base image name.

   Default is `informatica1041:1.0`, where `1.0` is the tag name.

4. Enter the working directory that has 100 GB of free disk space.
5. Applies for Enterprise Data Catalog deployed in an embedded cluster. Enter the path to the directory where the Apache Ambari binaries are located.
6. Optional. Enter the path to the directory where the Enterprise Data Catalog scanner binaries are located.
7. Click **Build**.

Step 3. Verify the Informatica Docker Image

You can verify that the Informatica image is present in the host specified while building the docker image.

To verify that the Informatica docker image exists, enter the docker images command from the command prompt. Ensure that the values for the tag, image ID, created date, and size information appears for the Informatica docker image.

The following sample displays the result of the docker image command:

```
[root@master abc]#docker images
REPOSITORY                    TAG         IMAGE ID      CREATED       SIZE
informatica1041pcrspostgre    1.0         f9923fb2cfa0  20 hours ago  17.7 GB
```

Run the Informatica Docker Image to Create a Domain

You can install Informatica products by running the docker image. Run the docker image to create nodes in the Informatica domain.

The first time that you run the docker image, choose to create a domain. When you create a domain, the node that you create becomes a gateway node in the domain. The gateway node contains a Service Manager that manages all the domain operations.

Step 1. Choose the Deployment Type

On the **Deployment type** page, you must select the deployment type and host instance to run the docker image.

1. Launch the Informatica container utility with the `sh startup.sh` command. Default port is 12386.

   To specify the startup command with a different port number, enter the values in the following format:

   ```
   sh startup.sh <required port number>
   ```
2. To run the Informatica Docker image, select **Run the docker image**.

   The **Deployment Type** page appears.
3. Enter the information for the deployment type.
   - Select Enterprise Data Preparation for the product.
   - Select **Existing Cluster** or **Embedded Cluster** based on the Enterprise Data Catalog deployment configuration.
   - Choose to host on Amazon Web Services (AWS), Microsoft Azure, or on-premises.
   - Select the Docker deployment type.
4. Optionally, you can choose to provide secure authentication to deploy the containers on the remote host. If you do not want to set the secure authentication for the containers, go to step 6.
5. If you enable secure authentication, set the authentication type to **Password** or **Key**.
   - For password type authentication, enter the machine IP address or host name, user name, password, and port to connect to the host machine.

- For key type authentication, enter the machine IP address or host name, user name, path of the host authentication key, and port number.

6. Click **Next**.

   The **Docker information** page appears.

Step 2. Provide Docker Information

On the **Docker information** page, provide the required information to connect to the docker image and container.

1. In the **Docker configuration** section, enter the docker image name, container name, path and file name of the license key, and the volume directory. Enter the existing directory to create tmpfs volume.

   Ensure that you add different values for the license key and volume directory.

2. Optionally, choose to expose additional ports in the container, and enter them as comma separated values.
3. Enter the network name.

   You can update the **Domain selection** section on the same page.

Step 3: Configure the Domain Configuration Repository

On the **Domain database** page, provide the domain configuration repository database information, JDBC connection information, and specify information for the domain and gateway node.

1. Select Oracle, Microsoft SQL Server, or PostgreSQL database to use for the domain configuration repository.
2. Optionally, specify if you created a database in the container. If you created a database in the container, complete the container configuration for the domain configuration repository.
3. Enter the database user account and password.
4. Specify whether the database is secure.

   If you select the secure database option, you need to provide inputs for the qualified path to the database truststore file, and truststore password.

5. Enter the parameters for the database.

   a. If you select Microsoft SQL Server or PostgreSQL, enter the schema name for the database. If you select Oracle, the installer creates the tables in the default schema.

   b. To enter the JDBC connection information, you can use either the JDBC URL or the JDBC connection string.

      To use the JDBC URL information, select **JDBC URL**, and specify the following JDBC URL properties:

      - **Database address.** Host name and port number for the database.
      - **Database service name.** Oracle service name, the database name for Microsoft SQL Server, or the database name for PostgreSQL.

      Optionally, choose to include additional JDBC parameters.

c. To use a custom JDBC connection string, select **JDBC Connection String**.

You can use the following syntax in the JDBC connection string to connect to a secure database:
Oracle

```
jdbc:Informatica:oracle://<host name>:<port number>;ServiceName=<service
name>;EncryptionMethod=SSL;HostNameInCertificate=<database host
name>;ValidateServerCertificate=<true or false>
```

Microsoft SQL Server

```
jdbc:Informatica:sqlserver://<host name>:<port
number>;SelectMethod=cursor;DatabaseName=<database
name>;EncryptionMethod=SSL;HostNameInCertificate=<database host
name>;ValidateServerCertificate=<true or false>
```

PostgreSQL

```
jdbc:Informatica:postgresql://<host name>:<port number>;DatabaseName=<database
name>;EncryptionMethod=SSL;HostNameInCertificate=<database host
name>;ValidateServerCertificate=<true or false>
```

6. In the **Domain and node configuration** section, enter the keyword and directory for the encryption key for the Informatica domain.
   - **Encryption key directory.** Directory in which to store the encryption key for the Informatica domain. The encryption key is created in the following directory: `<Informatica installation directory>/isp/config/keys`.
   - **Keyword.** Keyword to use to create a custom encryption key to secure sensitive data in the domain. The keyword cannot contain spaces; must be 8 to 20 characters long; and must include at least one number and one uppercase and lowercase letter.
7. Specify the domain name, node name, domain port number, domain user name, and password. When you enter the information for the domain and the gateway node that you want to create, consider the following guidelines:
   - The default domain name is Domain. The domain name must not exceed 128 characters and must be 7-bit ASCII only. The name cannot contain a space or any of the following characters: `` ` % * + ; " ? , < > \ / ``
   - The default port number for the node is 6005.
   - Consider the following guidelines when you create the domain user name:
     - The domain user name is the user for the domain administrator.
     - The name is not case sensitive and cannot exceed 128 characters.
     - The name cannot include a tab, newline character, or the following special characters: `% * + / ? ; < >`
     - The name can include an ASCII space character except for the first and last character.
   - The password for the domain administrator must be more than 2 characters and must not exceed 128 characters.
8. You can choose to configure application services or run the docker image.

   Note: If you choose to create the supported application services, you must also provide the related database information as part of the Informatica container utility tasks. You can also choose to set the cluster configuration. Make sure that you select the Model Repository Service, Data Integration Service, Cluster configuration, Catalog Service, and Enterprise Data Preparation Service.

Step 4: Create the Model Repository Service

On the **Model Repository Service** page, you can configure the Model repository database properties.

1. In the **Model Repository Service information** section, enter the database type, user ID, and user password. Specify the database name or database service name based on the database associated with Model repository.
2. Optionally, specify if you created a database in the container. If you created a database in the container, complete the container configuration for the Model repository.
3. Specify whether the database is secure.

   If you select the secure database option, you need to provide inputs for the qualified path to the database truststore file, truststore password, and specify the secure JDBC Parameters. By default, the value for the secure JDBC parameters is

   ```
   EncryptionMethod=SSL;HostNameInCertificate=;ValidateServerCertificate=false;
   ```
4. Enter the parameters for the database.
   a. If you select Microsoft SQL Server or PostgreSQL, enter the schema name for the database. If you select Oracle, the installer creates the tables in the default schema.
   b. To enter the JDBC connection information, you can use either the JDBC URL or the JDBC connection string.

      To use the JDBC URL information, select **JDBC URL**. To enter the connection information using the JDBC URL information, specify the following JDBC URL properties:

      - **Database address.** Host name and port number for the database.
      - **Database service name.** Oracle service name, the database name for Microsoft SQL Server, or the database name for PostgreSQL.

      Optionally, chose to include additional JDBC parameters.
   c. To use a custom JDBC connection string, select **JDBC connection string**.

      You can use the following syntax in the JDBC connection string to connect to a secure database:
      Oracle

      ```
      jdbc:Informatica:oracle://<host name>:<port number>;ServiceName=<service name>
      ```

      Microsoft SQL Server

      ```
      jdbc:Informatica:sqlserver://<host name>:<port
      number>;SelectMethod=cursor;DatabaseName=<database name>
      ```

      PostgreSQL

      ```
      jdbc:Informatica:postgresql://<host name>:<port number>;DatabaseName=<database name>
      ```
5. Click next to configure additional services, or run the docker image.

Step 5: Create the Data Integration Service

After you configure the Model Repository database, you can configure the service parameters for the application service.

1. In the **Data Integration Service information** section, you can specify the connection details for the Data Integration Service:
   a. Choose if you want to enable secure communication for the Data Integration Service with the HTTP, HTTPS, or HTTP&HTTPS connection options and specify the port number for the Data Integration Service.

b. If you enabled secure connection or if you enabled HTTPS connection, enter the HTTPS port number and the keystore file information. Choose to use a default keystore or a custom keystore file.

- If you use the default keystore, the installer creates a self-signed keystore file named Default.keystore in the following location: `<Informatica installation directory>/tomcat/conf/`
- If you use a custom keystore, specify the location and password.

2. Specify whether to use the default Informatica SSL certificates or to use your SSL certificates to secure the communication to the Data Integration Service.
3. If you provide the SSL certificate, specify the location and passwords of the keystore and truststore files. The directories must contain a keystore file named infa_keystore.jks or infa_keystore.pem and a truststore file named infa_truststore.jks or infa_truststore.pem.
4. Click next to configure additional services.

Step 6: Create the Cluster Configuration

Optionally, on the **Cluster configuration** page, you can specify the cluster configuration for the Hadoop environment.

1. In the **Cluster configuration information** section, choose to enter the name of the cluster configuration to create.
2. Specify the Hadoop distribution for the cluster.
3. Select the method to import the configuration properties from the Hadoop cluster to create the cluster configuration. You can import the properties from an archive file or directly from the cluster. For more information, see Cluster Configuration.

   - If you import from the archive file, enter the archive file and the cluster name.

     Note: If you select MapR or Amazon EMR cluster, you must import the cluster configuration properties from an archive file.

   - If you import the properties directly from the cluster, specify the host name or IP address of the cluster manager. You also need to provide the cluster manager port, user name, password, and the name of the cluster. Use the display name if the cluster manager manages multiple clusters. If you do not provide a cluster name, the wizard imports information based on the default cluster.

4. Optionally, choose to create the Hadoop, HDFS, Hive, and HBase connections.

   The installer appends the connection type to the cluster connection name to create each connection name.

Step 7: Create the Catalog Service

On the **Catalog Service** page, you can configure the Catalog Service properties.

1. If you choose to deploy Enterprise Data Catalog in an existing cluster, select **Yes** if you want the installer to automatically configure the ZooKeeper, HDFS, and YARN configuration properties or **No** if you want to configure the properties manually.

   If you choose **Yes**, select one of the following options to specify the Hadoop distribution:

   - Cloudera Manager
   - Hortonworks
   - HDInsight

Configure the following properties in the **Catalog Service Parameters** section:

Option	Description
Hadoop Cluster distribution URL	URL to access the Hadoop distribution.
Cluster URL user name	User name to access the Hadoop cluster.
Hadoop Cluster distribution URL password	Password to access the Hadoop cluster.

If you choose **No**, specify if the cluster is enabled for high availability and configure the following properties:

Option	Description
YARN resource manager URI	The service within Hadoop that submits the MapReduce tasks to specific nodes in the cluster. Use the following format: <host name>:<port>. <host name> represents the name or IP address of the YARN resource manager.. <port> represents the port number on which the YARN resource manager listens for Remote Procedure Calls (RPC).
YARN resource manager HTTP URI	The HTTP URI value for the YARN resource manager.
YARN resource manager scheduler URI	Scheduler URI value for the YARN resource manager.
ZooKeeper addresses	Multiple ZooKeeper addresses in a comma-separated list.
HDFS Namenode URI	The URI to access HDFS. Use the following format to specify the Namenode URI: <host name>:<port>. <host name> represents the host name or IP address of the Namenode. <port> represents the port number on which the Namenode listens for Remote Procedure Calls (RPC).
History Server HTTP URI	Specify a value to generate YARN allocation log files for resources. The Catalog Administrator displays the log URL as part of task monitoring.

2. Select **Yes** if the cluster is enabled for Kerberos authentication and configure the following properties in the **Catalog Service Parameters** section:

Option	Description
KDC domain name	Name of the KDC domain.
Keytab location	Location of the merged user and host keytab file.
Fully qualified path to the Kerberos configuration file	Fully qualified path to the Kerberos configuration file.
KDC host	Name of the KDC host machine.

3. Specify **Yes** if the cluster is enabled for SSL authentication and configure the following properties in the **Catalog Service Parameters** section:

Option	Description
Solr keystore file	Path to the Solr keystore file.
Solr keystore password	Password to access the Solr keystore file.
Truststore file	Path to the JKS file that includes all the merged cluster certificates.
Truststore password	Password to access the truststore file.

4. If you choose to deploy Enterprise Data Catalog in an embedded cluster, configure the following properties to create the Informatica Cluster Service in the **Informatica Cluster Service Creation** section:

Option	Description
Gateway user	User name to access the Apache Ambari Server or agent host. Make sure that the user name to access the Apache Ambari Server and the agent host are the same. Default is root.
Informatica Cluster Service name	Name of the Informatica Cluster Service. The name is not case-sensitive and must be unique within the domain. The name cannot exceed 128 characters or begin with @. The name cannot contain character spaces. The characters in the name must be compatible with the code page of the Model repository. The name cannot contain the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Informatica Hadoop cluster gateway port	Web port for the Apache Ambari Server.
Override default Hadoop Ambari password	Select **Yes** if you want to change the default password for the cluster.
KDC type	Select one of the following Kerberos Key Distribution Center (KDC) types if you had enabled Kerberos authentication for the cluster: • MIT KDC • Active Directory KDC
KDC host	Name of the KDC host machine.
Administrator Server host	The name of the administrator server machine that hosts the KDC server.
Fully qualified path to the Kerberos configuration file	Fully qualified path to the Kerberos configuration file.
Realm	Name of the Kerberos realm on the machine that hosts the KDC server.
Administrator Principal	The Kerberos administrator principal.
Administrator Password	The Kerberos administrator password.
Informatica Cluster Service HTTPS port	Port number for the HTTPS connection.

Option	Description
Enable secure communication for the service	Specify if you want to use the default keystore file generated by the installer or a custom keystore file to enable secure communication using SSL for the Informatica Cluster Service. If you select the option to provide a custom keystore file, you must provide the following details: • Informatica Cluster Service keystore file. Path and file name of the keystore file. • Informatica Cluster Service keystore alias. The keystore alias to access the keystore. • Informatica Cluster Service keystore password. Password to access the keystore file.
Directory to store HDFS, YARN, and ZooKeeper log files and data directories	If you want to specify a custom directory to save the HDFS, YARN, and ZooKeeper log files and data directories, select **Yes** and specify the path to the directory.

5. In the **Catalog Service Parameters** section, configure the following properties:

Option	Description
Catalog Service name	Name of the Catalog Service. The name is not case-sensitive and must be unique within the domain. The name cannot exceed 128 characters or begin with @. The name cannot contain character spaces. The characters in the name must be compatible with the code page of the Model repository that you associate with the Catalog Service. The name cannot contain the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Cluster Hadoop distribution URL*	URL to access the Hadoop distribution.
Cluster URL user name*	User name to access the Hadoop cluster.
Cluster Hadoop distribution URL password*	Password to access the Hadoop cluster.
Service cluster name*	Name of the service cluster.
Catalog Service HTTPS port	HTTPS port number that you want to configure for the Catalog Service.
Enable secure communication for the service	Specify if you want to use the default keystore file generated by the installer or a custom keystore file to enable secure communication using SSL for the Catalog Service. If you select the option to provide a custom keystore file, you must provide the following details: • Keystore file. Path and file name of the keystore file. • Keystore alias. The keystore alias to access the keystore. • Keystore password. Password to access the keystore file.
YARN queue name*	The YARN scheduler queue name that specifies the available resources on a cluster.

Option	Description
Slider ports	Specify if you want to configure slider ports for the Catalog Service. Configuring slider ports allows you to specify a range of ports that you want the Solr and HBase applications to use. Specify the values for the following properties if you want to configure slider ports: • Slider min port. The minimum slider port value that you want to configure for the Catalog Service. • Slider max port. The maximum slider port value that you want to configure for the Catalog Service.
Load type	Specify the size of data that you plan to load in the catalog. You can select from the following options: • demo • low • medium • high For more information about the size of data that you can load in the catalog, see the Informatica Installation for Enterprise Data Catalog guide.
Enable asset change email notifications*	Select **Yes** if you want to receive email notifications about the status of the Catalog Service.

* denotes the properties that are not applicable when you deploy Enterprise Data Catalog in an embedded cluster.

Note: When the cluster is Kerberos or SSL-enabled, you have to provide the Kerberos, keystore, and truststore files in the host where the domain is installed.

Step 8: Configure the Data Preparation Repository

On the **Data Preparation repository** page, configure the properties for the Data Preparation repository.

1. To associate application services with Enterprise Data Preparation, you can either create a new or use the existing Model Repository and Data Integration service.

   If you choose to create application services, go to configure the following services:

   - "Step 9: Create the Model Repository Service for Enterprise Data Preparation" on page 18
   - "Step 10: Create the Data Integration Service for Enterprise Data Preparation" on page 19
2. Select the database type for Data Preparation repository.

Option	Description
Oracle	Enter the database user name and password.
MySQL	Enter the database name, host name, port number, user name, and password.

3. Enter the parameters for the Oracle database.
   a. If you select Oracle, the installer creates the tables in the default schema.
   b. Specify whether the database is secure. If you select the secure database option, you need to provide inputs for the qualified path to the database truststore file, and truststore password.

c. To enter the JDBC connection information, you can use either the JDBC URL, secure JDBC parameters, or use the custom JDBC connection string.

To use the JDBC URL information, select **JDBC URL**, and specify the following JDBC URL properties:

- **Database address**. Host name and port number for the database.
- **Database service name**. Oracle service name.

Optionally, choose to include additional JDBC parameters.

d. To use a custom JDBC connection string, select **JDBC Connection String**.

You can use the following syntax in the JDBC connection string to connect to a secure database:
Oracle

```
jdbc:Informatica:oracle://<host name>:<port number>;ServiceName=<service
name>;EncryptionMethod=SSL;HostNameInCertificate=<database host
name>;ValidateServerCertificate=<true or false>
```

4. Enter the parameters for the MySQL database.
   a. To enter the JDBC connection information, you can use the JDBC connection string.
   b. To use a custom JDBC connection string, select **JDBC Connection String**.
   c. Specify whether the database is secure.

   If you select the secure database option, you need to provide inputs for the secure JDBC Parameters. By default, the value for the secure JDBC parameters is:

   ```
   EncryptionMethod=SSL;HostNameInCertificate=;ValidateServerCertificate=false;
   ```

5. Click next to configure the additional services.

Step 9: Create the Model Repository Service for Enterprise Data Preparation

On the **Model Repository Service for Enterprise Data Preparation** page, provide the information to create the Model Repository Service.

1. In the **Model Repository Service information** section, enter the database type, user ID, and user password. Specify the database name or database service name based on the database associated with Model repository.
2. Optionally, specify if you created a database in the container. If you created a database in the container, complete the container configuration for the Model repository.
3. Specify whether the database is secure.

   If you select the secure database option, you need to provide inputs for the qualified path to the database truststore file, truststore password, and specify the secure JDBC Parameters. By default, the value for the secure JDBC parameters is

   ```
   EncryptionMethod=SSL;HostNameInCertificate=;ValidateServerCertificate=false;
   ```

4. Enter the parameters for the database.
   a. If you select Microsoft SQL Server or PostgreSQL, enter the schema name for the database. If you select Oracle, the installer creates the tables in the default schema.
   b. To enter the JDBC connection information, you can use either the JDBC URL or the JDBC connection string.

To use the JDBC URL information, select **JDBC URL**. To enter the connection information using the JDBC URL information, specify the following JDBC URL properties:

- **Database address.** Host name and port number for the database.
- **Database service name.** Oracle service name, the database name for Microsoft SQL Server, or the database name for PostgreSQL.

Optionally, chose to include additional JDBC parameters.

c. To use a custom JDBC connection string, select **JDBC connection string**.

You can use the following syntax in the JDBC connection string to connect to a secure database:

Oracle

```
jdbc:Informatica:oracle://<host name>:<port number>;ServiceName=<service name>
```

Microsoft SQL Server

```
jdbc:Informatica:sqlserver://<host name>:<port
number>;SelectMethod=cursor;DatabaseName=<database name>
```

PostgreSQL

```
jdbc:Informatica:postgresql://<host name>:<port number>;DatabaseName=<database name>
```

5. Click next to configure additional services.

Step 10: Create the Data Integration Service for Enterprise Data Preparation

On the **Data Integration Service for Enterprise Data Preparation** page, provide the information to create the Data Integration Service.

1. In the **Data Integration Service information** section, you can specify the connection details for the Data Integration Service:
   a. Choose if you want to enable secure communication for the Data Integration Service with the HTTP, HTTPS, or HTTP&HTTPS connection options and specify the port number for the Data Integration Service.
   b. If you enabled secure connection or if you enabled HTTPS connection, enter the HTTPS port number.
2. Specify whether to use the default Informatica SSL certificates or to use your SSL certificates to secure the communication to the Data Integration Service.
3. If you provide the SSL certificate, specify the location and passwords of the keystore and truststore files. The directories must contain a keystore file named infa_keystore.jks or infa_keystore.pem and a truststore file named infa_truststore.jks or infa_truststore.pem.
4. Click next to configure additional services.

Step 11: Create the Interactive Data Preparation Service

On the **Interactive Data Preparation Service** page, provide the required information to create the Interactive Data Preparation Service.

1. Enter a name for the Interactive Data Preparation Service.
2. To configure Enterprise Data Preparation with rules, enter the Model Repository Service and Data Integration Service name.
3. Enter the node name where you want to run the Interactive Data Preparation Service.
4. Choose if you want to enable secure communication for Interactive Data Preparation Service. Specify the HTTP or HTTPS port number based on your option.

5. Specify whether to use the default Informatica SSL certificates or to use your SSL certificates to secure the communication to the Interactive Data Preparation Service.
6. If you want to provide the custom SSL certificate, specify the location and passwords of the keystore and truststore files. The directories must contain a keystore file named infa_keystore.jks or infa_keystore.pem and a truststore file named infa_truststore.jks or infa_truststore.pem.
7. Choose the Hadoop authentication mode. Specify the HDFS storage location, connection, and local storage location.

   Specify the following details if you choose the Kerberos authentication mode:

Option	Description
HDFS principal name	Enter the HDFS principal name. Principal name consists of a primary and an instance name.
Hadoop impersonation user name	Enter the user impersonation name for the Hadoop environment.
Kerberos keytab file	Enter the host keytab file name. **Note:** Available in the cluster. You must copy the file to the domain host.
Fully qualified path to the Kerberos configuration file	Enter the fully qualified path to the Kerberos configuration file. **Note:** Available in the cluster. You must copy the file to the domain host.
KDC host	Enter the IP address of the KDC host machine.

8. Enter the path to the JKS file that includes all the merged cluster certificates for the truststore file and provide password to access that file, if the cluster is SSL enabled.
9. Choose to enable the Interactive Data Preparation Service immediately after its creation.

Step 12: Create the Enterprise Data Preparation Service

On the **Enterprise Data Preparation Service** page, provide the required information to create the Enterprise Data Preparation Service.

1. Enter the Enterprise Data Preparation Service name.
2. Enter the Interactive Data Preparation Service name to associate with the Enterprise Data Preparation Service.
3. Enter the Model Repository Service name to associate with the Enterprise Data Preparation Service.
4. Enter the Data Integration Service name to associate with the Enterprise Data Preparation Service.
5. Enter the node name where you want to run the Enterprise Data Preparation Service.
6. Choose if you want to enable secure communication for Enterprise Data Preparation Service. Specify the HTTP or HTTPS port number based on your option.
7. Specify whether to use the default Informatica SSL certificates or to use your SSL certificates to secure the communication to the Interactive Data Preparation Service.
8. If you want to provide the custom SSL certificate, specify the location and passwords of the keystore and truststore files. The directories must contain a keystore file named infa_keystore.jks or infa_keystore.pem and a truststore file named infa_truststore.jks or infa_truststore.pem.

9. Provide the Data Preparation connection information:

Option	Description
HDFS connection	Enter the HDFS connection name.
HDFS working directory	Location of the HDFS working directory.
Hadoop connection	Enter the Hadoop connection name.
Hive connection	Enter the Hive connection name.
Hive table storage format	Choose one of the following storage formats: • Default format • Parquet • ORC
Local system directory	Location of the local system directory.

10. Choose the Hadoop authentication mode.

    Specify the following details if you choose the Kerberos authentication mode:

Option	Description
EDP Kerberos Principal	Enter the Enterprise Data Preparation Kerberos principal name. Principal name consists of a primary and an instance name.
EDP Kerberos keytab file	Enter the Enterprise Data Preparation Kerberos keytab file name.

11. Choose to enable the Enterprise Data Preparation Service immediately after its creation.

Run the Informatica Docker Image to Join a Domain

After you create the domain, you can run the Informatica container utility on another machine to join the domain. When you join the domain, you need to create a node on the current machine and then add the node to the domain. The Informatica container utility configures the cluster related files, that you configured during domain creation, on the node where you want to join.

Note: If you selected persistent volume, you cannot join the domain.

Step 1. Choose the Deployment Type

On the **Deployment type** page, you must select the deployment type and host instance to run the docker image.

1. Launch the Informatica container utility with the `sh startup.sh` command. Default port is 12386.

   To specify the startup command with a different port number, enter the values in the following format:

   ```
   sh startup.sh <required port number>
   ```

2. To run the Informatica Docker image, select **Run the docker image** in the Informatica container utility.

   The **Deployment type** page appears.

3. Enter the information for the deployment type selection.
   - Select Enterprise Data Preparation for the product.
   - Choose to host on Amazon Web Services (AWS), Microsoft Azure, or on-premises.
   - Select the Docker deployment type.
4. Optionally, you can choose to provide secure authentication to deploy the containers on the remote host. If you do not want to set the secure authentication for the containers, go to step 6.
5. If you enable secure authentication, set the authentication type to **Password** or **Key**.
   - For password type authentication, enter the machine IP address or host name, user name, password, and port to connect to the host machine.
   - For key type authentication, enter the machine IP address or host name, user name, path of the host authentication key, and port number.
6. Click **Next**.

   The **Docker information** page appears.

Step 2. Provide Docker Information

On the **Docker information** page, provide the required information to connect to the docker image and container.

1. In the **Docker configuration** section, enter the docker image name, container name, path and file name of the license key, and the volume directory. Enter the existing directory to create tmpfs volume.

   Ensure that you enter different values for the volume directory and the license key directory.
2. Optionally, choose to expose additional ports in the container, and enter them as comma separated values.
3. Enter the network name.

   You can update the **Domain selection** section on the same page.

Step 3. Provide the Domain Connection Information

On the **Docker Details** page, provide information to join a domain, connection details for Informatica Administrator, and whether to secure communication to the domain or not.

1. In the **Domain Selection** section, choose to join a domain.

   The installer creates a node on the container where you install. You can specify the type of node to create and the domain to join.
2. Optionally, choose to enable secure communication for services in the Informatica domain.

   By default, if you enable secure communication for the domain, the installer sets up an HTTPS connection. You can also create a domain configuration repository on a secure database.
3. Optionally, choose the type of node you want to create. Default is worker node. You can also enable a secure HTTPS connection to Informatica Administrator.
4. Specify the connection details for Informatica Administrator.
   a. If you disabled secure communication for the domain, you can specify whether to set up a secure HTTPS connection for Informatica Administrator.

b. If you enabled secure connection for the domain or if you enabled HTTPS connection, enter the HTTPS port number and the keystore file information. Choose to use a default keystore or a custom keystore file.

- If you use the default keystore, the installer creates a self-signed keystore file named Default.keystore in the following location: `<Informatica installation directory>/tomcat/conf/`
- If you use a custom keystore, specify the location and password.

5. Specify whether to use the default Informatica SSL certificates or to use your SSL certificates to secure domain communication.
6. If you provide the SSL certificate, specify the location and passwords of the keystore and truststore files. The directories must contain a keystore file named infa_keystore.jks or infa_keystore.pem and a truststore file named infa_truststore.jks or infa_truststore.pem.
7. Click **Next**.

   The **Domain Configuration** page appears.

Step 4. Configure the Domain

On the **Domain Configuration** page, provide the domain, node, gateway node, and gateway container information.

1. In the **Domain Configuration** section, specify the domain name, gateway container name, gateway node port, domain user name, and password. When you enter the information for the domain and the gateway node that you want to create, consider the following guidelines:
   - The default domain name is Domain. The domain name must not exceed 128 characters and must be 7-bit ASCII only. The name cannot contain a space or any of the following characters: `` ` % * + ; " ? , < > \ / ``
   - The gateway container name is container name of the master gateway node.
   - Consider the following guidelines for the domain user name:
     - The domain user name is the user for the domain administrator.
     - The name is not case sensitive and cannot exceed 128 characters.
     - The name cannot include a tab, newline character, or the following special characters: `% * + / ? ; < >`
     - The name can include an ASCII space character except for the first and last character.
   - The password for the domain administrator must be more than 2 characters and must not exceed 128 characters.
2. Optionally, choose to join an Informatica domain created through the container database.
3. In the **Domain Configuration** section, enter the key and directory for the encryption key for the Informatica domain.
   - **Select the encryption key.** Specify the custom site key to secure sensitive data in the domain. Before you can enter the site key path, copy the file from the encryption key directory within the container to the host with the following command: docker cp <container name>: <encryption key directory>/siteKey <dst path in host>
   - **Encryption key directory.** Directory in which to store the encryption key for the Informatica domain. The encryption key is created in the following directory: `<Informatica installation directory>/isp/config/keys`.
4. Under the **Domain Configuration** section, enter the node host name, node name, and node port number for the current node. The node host name is the container name of the node.
5. Click **Deploy**.

   The docker image runs.

6. To verify that the container is present in the host specified, run the docker ps -a command from the command prompt.

   Ensure that you can see the container ID, image, names, command, created date, and status of each container appears.
   The following sample displays the result of the docker ps -a command:

```
[root@irlcmg08 source]#docker ps -a
CONTAINER ID PORTS  IMAGE NAMES                          COMMAND
CREATED     STATUS
566c418d0972        informaticaltd/pcdqservices:1041pc  "/Installer/launcher..." 2 days
ago  Up 2 days
```

7. To verify the docker images, run the docker images -a command in the command line.

   Ensure that you can see the repository name, tags, and the size information for the images.

After the docker image run completes, you can view the post-installation summary. You can view the installation log files to get more information about the tasks performed by the utility. The installation is complete when the Informatica_<Version>_Services_<timestamp>.log file is created in the installation directory.

Complete the Post-Install Tasks

After you run the Informatica container utility, perform the post-installation tasks.

Complete the Domain Configuration

To complete the domain configuration after you install the Informatica services, perform the following tasks:

- Verify locale settings and code page
- Configure Informatica environment variables
- Configure the library path environment variables
- Configure locale environment variables

Create the Application Services

Complete the following tasks:

Update the hosts file to access the Administrator tool through the browser.

Enter the following information in the hosts file to access the Administrator tool through the browser:

`<IP address of the machine> <host name of Informatica domain>.`

You can find the hosts file in the following location: `/etc/hosts`.

Log in to Informatica Administrator.

You must have a user account to log in to the Informatica Administrator web application with the administrator name that you used in the wizard. You can access the Administrator tool using the following link format: `https://<host name of the machine>:<port number>/administrator`

Create the application services.

If you did not create the Model Repository Service, Data Integration Service, Catalog Service, and the Informatica Cluster Service when you created the domain, you can use the service creation wizard in Administrator tool to create the application services. You need to create the Informatica Cluster Service if you are deploying Enterprise Data Catalog in an embedded cluster.

Create the following services:

1. Model Repository Service
2. Data Integration Service
3. Interactive Data Preparation Service
4. Enterprise Data Preparation Service
5. Catalog Service

Starting and Stopping the Informatica Services

Run infaservice.sh to start and stop the Informatica daemon.

You can start the daemon with the infaservice.sh startup command. To stop the daemon, enter the infaservice.sh shutdown command. By default, infaservice.sh is installed in the following directory:

```
<Informatica installation directory>/tomcat/bin
```

Note: If you use a softlink to specify the location of infaservice.sh, set the INFA_HOME environment variable to the location of the Informatica installation directory.

Troubleshooting Docker

I tried to build the docker image, but the build image fails.

If build image fails during installer tar file extraction or before the docker build image starts, verify that adequate disk space is available in the current working directory. Enterprise Data Preparation requires 100 GB disk space in the current working directory.

If the build image fails at the step `Copy/Binaries/home/Informatica` in the log file, then there is not enough disk space in the docker configuration directory. To find the docker configuration directory, run the docker info command. Enterprise Data Preparation requires 56 GB disk space in the docker configuration directory.

You can also free up some space from the docker configuration directory when you delete a dangling image. When you create the new build of the image but do not specify a new name, you create a dangling image and the old image becomes the dangling image. Those old images remain untagged and display "<none>" as the name when you run the docker images command.

To delete the dangling images, complete the following steps:

1. To list all dangling images and to get the image ID, run the command: docker images -f dangling=true
2. To delete each dangling image, run the command: docker image rm -f <image id>

If the build image fails with the following error, you need to manually pull the docker image:

```
Sending build context to Docker daemon 25.75GB
Step 1/30: FROM registry.access.redhat.com/rhel7:7.7
Trying to pull repository registry.access.redhat.com/rhel7 ...
Installation Status:ERROR
```

To fix the issue, you can manually run the following command:

```
docker pull registry.access.redhat.com/rhel7:7.7
```

If the build image fails with the following error, the docker build script has failed to access the repository to download dependencies:

```
This system is not registered with an entitlement server.
```

To register and subscribe the machine where you build the docker image:

1. Run the yum repolist all command to see all the repositories.
2. Use the subscription-manager to register with the following command: subscription-manager repos --enable.
3. To enable custom repositories, run the command: yum-config-manager --enable

If the build image displays the authenticationrequired error when you run the docker pull command to pull the image from RHEL, the authentication with the docker authentication might have expired. Run the docker logout command and then pull the docker image. If the issue persists, try to pull the docker image from registry.redhat.io instead of registry.access.redhat.com. If any issue remains with RHEL, you could try to use Centos7 for proof of concept cases.

I cannot access Administrator tool from the browser for Docker

Check the IP address from where you access the Administrator tool:

Ensure that you use the host IP of the node on which container is running to access the Administrator tool.

Appendix A: Sample Silent Properties for CI/CD for Build Image for Axon

The following example shows the contents of the file:

```
DOWNLOAD_AND_RUN=0
HubUserName=
IMAGE_NAME=testaxon:1.0
HubPassword=
OS_NAME=registry.access.redhat.com/rhel7:7.8
AGENT_VERSION=5
INFA_REPO=0
LICENSE_KEY_LOC=/data2/EDC_Resources/abdeic_1020_All_8850.key
PRIVATE_HUB=1
INSTALL_OPTION=1
AMBARI_FILE_LOC=/data2/1040_Binaries/ambaribinariesredhat7.tar.gz
INSTALL_DPM=0
SCANNER_FILE_LOC=/data2/1040_Binaries/ScannerBinaries.zip
DPM_TAR_FILE_LOC=/data2/1040_Binaries/10.4.0_DataPrivacyManagement_Installer_linux-
x64_Build_GA.tar
productType=AXON
TAR_FILE_LOC=/misc/infastore/Axon/7.1.0/pre-prod/aio-development/4594/Axon_7.1.0_linux-x64.tar
```

Appendix B: Sample Silent Properties for CI/CD for Run Image for Axon

The following example shows the contents of the file:

```
DIS_TRUSTSTORE_FILE_PATH=
CLOUD_SUPPORT_ENABLE=0
MRS_SERVICE_NAME=
ENABLE_KERBEROS=0
DIS_DOMAIN_HOST=inaxonqa001
ADVANCE_CONFIG_PAGE_STATUS=1
MRS_USER_NAME=Administrator
APP_HTTP_PORT=9001
UPG_BACKUP_DIR=/root/Informatica/DQAGENT_HOME
DIS_SERVICE_NAME=
DIS_SSL_CUSTOM_PASSWD=
SAML_AUTHENTICATION=false
INSTALL_TYPE=0
MRS_USER_PASSWD=Administrator
ADVANCE_PORT_CONFIG=0
MRS_IS_SSL_ENABLED=0
```

```
AXON_HTTPS_ENABLED=0
DIS_IS_SSL_ENABLED=0
AXON_AGENT_VERSION=1
USER_INSTALL_DIR=/home/Informatica
DIS_USER_NAME=Administrator
MRS_IS_SSL_CUSTOM=0
AXON_USERNAME=admin@informatica.com
RULE_AUTOMATION_ENABLED=0
MRS_NODE_NAME=Node01
DIS_IS_SSL_CUSTOM=0
MRS_DOMAIN_HOST=inaxonqa001
USER=
DOMAIN_NAME=Domain
UPGRADE_WITHOUT_PC=1
ENABLE_USAGE_COLLECTION=1
DIS_NODE_NAME=Node1
DIS_USER_PASSWD=Administrator
AXON_URL=http://10.65.145.214:9999
MRS_TRUSTSTORE_FILE_PATH=
DIS_NODE_PORT=6005
DIS_SECURITY_DOMAIN=1
MRS_SECURITY_DOMAIN=1
RULE_AUTOMATION_CONNECTION_NAME=
AXON_PASSWORD=Changeme@123
PASSWORD_COMPLEXITY=false
MRS_SSL_CUSTOM_PASSWD=
MRS_NODE_PORT=6005
```

Appendix C: Sample Silent Properties for CI/CD for Build Image for Axon Agent

The following example shows the contents of the file.

```
DOWNLOAD_AND_RUN=0
HubUserName=
IMAGE_NAME=testaxonagent:1.0
HubPassword=
OS_NAME=registry.access.redhat.com/rhel7:7.8
AGENT_VERSION=5
INFA_REPO=0
LICENSE_KEY_LOC=/data2/EDC_Resources/abdeic_1020_All_8850.key
PRIVATE_HUB=1
INSTALL_OPTION=1
AMBARI_FILE_LOC=/data2/1040_Binaries/ambaribinariesredhat7.tar.gz
INSTALL_DPM=0
SCANNER_FILE_LOC=/data2/1040_Binaries/ScannerBinaries.zip
DPM_TAR_FILE_LOC=/data2/1040_Binaries/10.4.0_DataPrivacyManagement_Installer_linux-
x64_Build_GA.tar
productType=AGENT
TAR_FILE_LOC=/misc/infastore/Axon/7.1.0/pre-prod/aio-
development/4594/Axon_7.1.0_linux-x64.tar
```

Appendix D: Sample Silent Properties for CI/CD for Run Image for Axon Agent

Enter a short description of the concept here (required).

```
CHANGE_REQUEST_PORT=5000
EIC_RESOURCE_NAME=
CLOUD_SUPPORT_ENABLE=0
ENABLE_KERBEROS=0
HTTPS_PORT=
MARKETPLACE_PWD=
UPG_BACKUP_DIR=/root/Informatica/Axon
DB_PWD=
EXTERNAL_EXSISTING_POSTGRES_CONNECT_STRING=
```

```
CHANGEREQUEST_DB_PWD=
VALUELIST_DB_PWD=
HTTPS_ENABLED=0
WORKFLOW_PORT=8081
ADVANCE_PORT_CONFIG=0
DOC_UPLOAD_DB_PWD=
BULK_UPLOAD_SERVICE_PORT=7000
BULK_UPLOAD_COMMITOR_PORT=7030
EXTERNAL_EXSISTING_POSTGRES_ADMIN_PASSWD=
VALUE_LIST_PORT=7060
EXTERNAL_EXSISTING_POSTGRES_STATUS=0
UNISON_PORT=7050
DOC_UPLOAD_PORT=4000
EIC_PORT=
SATS_USERNAME=
EXTERNAL_EXSISTING_POSTGRES_DIR=
UNISON_DB_PWD=
SEGMENTATION_PWD=
SSL_CERTIFICATE=
BULK_UPLOAD_VALIDATOR_PORT=7010
SEGMENTATION_PORT=7621
CUSTOM_DATABASE_PASSWD_STATUS=0
SATS_HOST=
EXTERNAL_POSTGRES_DIR=
EIC_HOST=
ADVANCE_CONFIG_PAGE_STATUS=1
SATS_CONFIGURATION_OPTION=0
SATS_PASSWORD=
MEMCACHE_PORT=11211
HTTP_PORT=9999
EIC_CONFIGURATION_OPTION=0
EIC_NAMESPACE=
EXTERNAL_EXSISTING_POSTGRES_PORT=
SAML_AUTHENTICATION=false
INSTALL_TYPE=0
SSL_KEY=
LICENSE_KEY_LOC=/data/axon/license.key
SATS_NAMESPACE=
MSG_BROKER_PORT=5672
INSTALL_OPTION=1
NOTIFICATION_DB_PWD=
MARKETPLACE_PORT=7704
SATS_PORT=
PROCESS_CONFIGURATION_PORT=2480
BULK_UPLOAD_PREVALIDATOR_PORT=7020
EIC_PASSWORD=
USER_INSTALL_DIR=/home/Informatica
PROCESS_CONFIGURATION_PORT2=2424
PROXY_HOST_NAME_OPTION=0
KEY_LIST_FILE_LOC=/home/Informatica/config/keyFile.txt
PROXY_HOST_NAME=
EXTERNAL_POSTGRES_DB_NAME=
EXTERNAL_POSTGRES_STATUS=0
BULKUPLOAD_DB_PWD=
USER=nroot
UPGRADE_WITHOUT_PC=1
ENABLE_USAGE_COLLECTION=1
EXTERNAL_POSTGRES_HOST=
EIC_USERNAME=
EXTERNAL_EXSISTING_POSTGRES_ADMIN_USER=
IS_POSTGRES_CONNECT_STRING=0
IS_LOGGING_ENABLED=0
NOTIFICATION_PORT=3000
EXTERNAL_POSTGRES_PASSWD=
PASSWORD_COMPLEXITY=false
```

Appendix A: Sample Silent Properties for CI/CD for Build Image

The following example shows the contents of the file:

```
DOWNLOAD_AND_RUN=0
HubUserName=
IMAGE_NAME=informatica1040:1.0
INSTALL_EDP=0
CLUSTER_TYPE=2
HubPassword=
OS_NAME=registry.access.redhat.com/rhel7:7.8
INFA_REPO=0
PRIVATE_HUB=1
INSTALL_OPTION=1
AMBARI_FILE_LOC=
INSTALL_DPM=0
SCANNER_FILE_LOC=
DPM_TAR_FILE_LOC=
productType=PC
TAR_FILE_LOC=/home/
```

Appendix B: Sample Silent Properties for CI/CD for Run Image

The following example shows the contents of the file:

```
ENABLE_KERBEROS=0
DIS_KEYTAB_FILELOC=
MRS_DB_CUSTOM_STRING=jdbc:informatica:oracle://host_name:port_no;ServiceName=
MRS_KEYTAB_FILELOC=
CMS_KEYSTORE_FILE=
DPS_KEYSTORE_PSSWD=
CMS_DATA_ACCESS_CONNECT_STRING=
OS_NAME=centos:centos7
HDFS_SERVICE_PRINCIPAL=
CREATE_CONNECTION=true
ADVANCE_PORT_CONFIG=0
PRIVATE_HUB=1
EDP_KERBEROS_KEYTAB_FILE=/etc/merge.keytab
EDP_DIS_CUSTOM_SELECTION=false
KSTORE_PSSWD=
EDP_DIS_KEYSTORE_PASSWD=
CLUSTER_PASSWORD=
CMS_DB_UNAME=CPI_CMS
WORKING_DIR=/home
DPS_TRUSTSTORE_DIR=
SPN_SHARE_LEVEL=
YARN_RESOURCE_MANAGER_HTTP_URI=
MAX_PORT=
EDC_KDC_HOST_IP=
DOMAIN_USER=Administrator
LOAD_DATA_DOMAIN=0
CONFIRM_HADOOP_AMBARI_PASSWD=
HADOOP_KEYSTORE_FILE=
DPS_HTTPS_PORT=
MRS_SERVICE_NAME_EDP=Model_Repository_Service_EDP
TRUSTSTORE_MRS_DB_FILE=
PWH_DB_PASSWD=CPI_PWH
YARN_QUEUE_NAME=
CMS_DB_CUSTOM_STRING=
PWH_DB_ADDRESS=insrh74dsg015.informatica.com:1521
PCRS_DB_HOST=
PWH_DATA_ACCESS_CONNECT_STRING=
EDP_DIS_KEYSTORE_DIR=
IS_CLUSTER_SSL_ENABLE=
EDP_MRS_SSL_DEFAULT_STRING=
HADOOP_IMPERSONATION_USER=
CATALOGUE_SERVICE_KEYSTORE_PASSWD=
EDP_SERVICE_NAME=Enterprise_Data_Preparation_Service
IMPORT_METHOD=2
```

```
USE_DB_CONTAINER=false
DPS_CONFIGURE_SERVICES=true
HDFS_SERVICE_NAME_HA=
CMS_SERVICE_NAME=CMS
EDP_PROTOCOL_TYPE=
TRUSTSTORE_DB_FILE=
YARN_RESOURCE_MANAGER_SCHEDULER_URI=
DPS_DB_HOST=
SERVICE_PROVIDER_ID=
TRUSTSTORE_MRS_DB_PASSWD=
PWH_DB_CUSTOM_STRING_SELECTION=0
DB_CUSTOM_STRING=jdbc:informatica:oracle://host_name:port_no;ServiceName=
CATALOGUE_SERVICE_KEYSTORE_FILE=
DIS_CUSTOM_SELECTION=false
INSTALLATION_ENVIRONMENT=Sandbox
INSTALL_TYPE=0
KSTORE_FILE_LOCATION=
LICENSE_KEY_LOC=/root/license.key
EDP_MRS_DB_CUSTOM_STRING_SELECTION=0
TRUSTSTORE_DB_PASSWD=
AMBARI_FILE_LOC=
MRS_DB_TYPE=Oracle
HADOOP_NODES=
AC_PORT=
MONITORING_SSL_DEFAULT_STRING=EncryptionMethod=SSL;HostNameInCertificate=;ValidateServerCertific
ate=false;
USER_REALM_NAME=
HADOOP_TLS_HTTPS_PORT=
KERBEROS_KEYTAB_FILE=/etc/merge.keytab
DIS_HTTPS_PORT=18095
INSTALL_DPM=0
CMS_DB_CUSTOM_STRING_SELECTION=0
PWH_DB_TYPE=Oracle
PCRS_DB_PSSWD=
DPS_CUSTOM_HTTPS_ENABLED=false
DPM_TAR_FILE_LOC=
NODE_TRUSTSTORE_DIR=
CUSTOM_HTTPS_ENABLED=0
LOAD_TYPE=low
USER_INSTALL_DIR=/home/Informatica
PCRS_DB_TYPE=Oracle
DPS_PROTOCOL_TYPE=
HADOOP_TRUSTSTORE_FILE=
DPM_LICENSE_KEY_LOC=/data2/EDC_Resources/DataPrivacyManagement.key
PWH_SQLSERVER_SCHEMA_NAME=
NEW_HADOOP_AMBARI_PASSWD=
DB_CONTAINER_NAME=
EDP_CUSTOM_HTTPS_ENABLED=false
CMS_KEYSTORE_PASSWD=
DPS_DB_USER=
HADOOP_GATEWAY_HOST=
SECURITY_DOMAIN_NAME=
MONITORING_KEYTAB_FILELOC=
MRS_DB_PASSWD=
DPS_DB_TRUSTSTORE_PASSWD=
EDP_TRUSTSTORE_PSSWD=
KRB5_FILE_LOCATION=/etc/krb5.conf
CMS_ADVANCE_JDBC_PARAM=
EDP_DIS_HTTPS_PORT=
VOLUME_DIRECTORY=/root/swathi/shared
MRS_ADVANCE_JDBC_PARAM=
EDP_DPS_SERVICE_NAME=Interactive_Data_Preparation_Service
AC_SHUTDWN_PORT=
CREATE_PCRS=false
SSL_ENABLED=false
DIS_SERVICE_NAME_EDP=Data_Integration_Service_EDP
LOCAL_STORAGE_DIR=/home/toolprod/satish/temp
CLOUD_SUPPORT_ENABLE=0
IS_SERVICE_NAME=
HTTPS_PORT=
CATALOGE_SERVICE_KEYTAB_LOCATION=
```

```
HDFS_HOST_NAME=
MONITORING_DB_ADDRESS=
DEFAULT_HTTPS_ENABLED=1
CLUSTER_HADOOP_DISTRIBUTION_TYPE=
EDL_KRB5_FILE_LOCATION=/etc/krb5.conf
DB_PASSWD=admin
EDP_MRS_DB_UNAME=
SCANNER_FILE_LOC=
SSH_HOST_NAMES=
NODE_KEYSTORE_PASSWD=
DPS_KEYSTORE_DIR=
CMS_DB_TYPE=Oracle
INFA_CONTAINER_NAME=infa
EDP_MRS_DB_ADDRESS=HostName:PortNumber
SAML_TRUSTSTORE_PASSWD=
DPS_HTTP_PORT=
DIS_PROTOCOL_TYPE=http
HDFS_SYSTEM_DIR=/datalake/EDL
EDP_MRS_SERVICE_NAME=Model_Repository_Service_EDP
HADOOP_CUSTOM_SELECTION=false
DPS_KDC_HOST_IP=
MIN_PORT=
CREATE_LAKE_SERVICES=0
SAML_TRUSTSTORE_ALIAS=informatica llc
EDP_DIS_SERVICE_NAME=Data_Integration_Service_EDP
CLUSTER_HADOOP_DISTRIBUTION_URL=
KEYTAB_LOCATION=
IHS_ADMINISTRATOR_PASSWORD=
DPS_CLUSTER_TRUSTSTORE_FILE_LOC=
SILENT_FILE_LOC=/root/appconTemp/informaticaltd/dei/SilentInput.properties
ENABLE_EDP_SERVICE=true
EDP_KERBEROS_PRINCIPAL=pam_user@PLATFORMKRB.COM
DOWNLOAD_AND_RUN=0
DB_SSL_ENABLED=false
SECURITY_MODE=Kerberos
EDP_MRS_DB_CUSTOM_STRING=
MRS_DB_SSL_ENABLED=false
DPS_DB_TRUSTSTORE_FILE=
EDP_DIS_TRUSTSTORE_PASSWD=
SAML_TRUSTSTORE_DIR=
CATALOGUE_SERVICE_NAME=catalog_service
CATALOGUE_SERVICE_SOLR_KEYSTORE_FILE=
CMS_SERVICE_NAME_ASSOCIATED=CMS
KEY_SRC_LOCATION=
TRUSTSTORE_MONITORING_DB_PASSWD=
CATALOGUE_SERVICE_PORT=8180
DISTRIBUTION_TYPE=cdh
NODE_TRUSTSTORE_PASSWD=
IDP_URL=info@informatica.com
HubUserName=
IHS_ADMINISTRATOR_PRINCIPAL=
CLUSTER_CONFIG_NAME=
CLUSTER_NAME=
DPS_NODE_NAME=Node01
MONITORING_DB_SSL_ENABLED=false
TRUSTSTORE_MONITORING_DB_FILE=
INSTALL_EDP=0
HubPassword=
ENABLE_USAGE_COLLECTION=1
UPGRADE_WITHOUT_PC=1
CONTAINER_DN=
INFA_REPO=0
LOCAL_SYSTEM_DIR=/home/toolprod/satish/temp
CREATE_DOMAIN=1
SQLSERVER_SCHEMA_NAME=
PWH_DB_SERVICENAME=SSL12CR2.informatica.com
CLUSTER_PORT=
CMS_HTTP_PORT=9050
SERVICE_CLUSTER_NAME=pam_user
HADOOP_KEYSTORE_PASSWD=
DPS_SSL_DEFAULT_STRING=
```

```
CATALOGUE_SERVICE_KEYSTORE_ALIAS=
IS_CLUSTER_HA_ENABLE=
DPS_DB_TYPE=
CMS_CUSTOM_SELECTION=false
EDP_TRUSTSTORE_MRS_DB_PASSWD=oracle4u
DOMAIN_PSSWD=admin
EDP_HTTPS_PORT=
MONITORING_DB_SERVICENAME=
DB_ADDRESS=oracle
DIS_SERVICE_NAME=Data_Integration_Service
HTTPS_ENABLED=0
SERVES_AS_GATEWAY=0
EDP_MRS_SQLSERVER_SCHEMA_NAME=
DOMAIN_HOST_NAME=infa
MONITORING_SQLSERVER_SCHEMA_NAME=
PCRS_DB_PORT=1521
NODE_KEYSTORE_DIR=
SERVER_PORT=
DOCKER_NETWORK_NAME=
OVERRIDE_HADOOP_AMBARI_PSD=false
KDC_CERT_FILE_LOC=
DIS_HTTP_PORT=
EDP_MRS_ADVANCE_JDBC_PARAM=
MRS_DB2_TABLESPACE=
EDP_DIS_PROTOCOL_TYPE=
DPS_DB_PORT=
EDP_MRS_DB2_TABLESPACE=
AUTH_MODE=Kerberos
NODE_NAME=node01
CLUSTER_HADOOP_DISTRIBUTION_URL_PASSWD=
DB2_TABLESPACE=
KERBEROS_CONF_FILE_LOC=/etc/krb5.conf
ENABLE_CUSTOM_SERVICE_PROVIDER_ID=0
KERBEROS_DOMAIN_PSSWD=
INSTALL_LDM=0
CMS_DB_SERVICENAME=SSL12CR2.informatica.com
MONITORING_DB_CUSTOM_STRING=
CMS_SQLSERVER_SCHEMA_NAME=
EDP_TRUSTSTORE_MRS_DB_FILE=/home/toolprod/INFA_Automation/newssldb/keystore_orassl.jks
HDFS_PRINCIPAL_NAME=nn/_HOST@PLATFORMKRB.COM
KEY_DEST_LOCATION=/home/Informatica/isp/config/keys
MRS_SERVICE_NAME=Model_Repository_Service
DB_UNAME=admin
ENABLE_CMS_SERVICE=true
CLUSTER_TYPE=2
EDP_DIS_TRUSTSTORE_DIR=
SERVICE_REALM_NAME=
KDC_KEYTAB_LOCATION=/etc/merge.keytab
SAML_AUTHENTICATION=false
MONITORING_DB_UNAME=
DIS_TRUSTSTORE_DIR=
CLUSTER_CONFIG_VALIDATION=0
HADOOP_GATEWAY_PORT=8080
HOST_SSH_USER=root
MRS_SQLSERVER_SCHEMA_NAME=
DOMAIN_PORT=6005
DB_SERVICENAME=ora
RESUME_INSTALLATION=false
EDP_TRUSTSTORE_DIR=
DPS_DB_CUSTOM_STRING_SELECTION=false
PWH_ADVANCE_JDBC_PARAM=
EDP_MRS_DB_SERVICENAME=DBServiceName
ENABLE_DB_SSL=false
DPS_DB_SCHEMA=
CONFIGURE_SERVICES=1
MONITORING_ADVANCE_JDBC_PARAM=
EDP_HTTP_PORT=
MRS_SSL_DEFAULT_STRING=EncryptionMethod=SSL;HostNameInCertificate=;ValidateServerCertificate=fal
se;
TUNING_SIZE=sandbox
DPS_TRUSTSTORE_PSSWD=
```

```
KDC_HOST=
DOMAIN_CNFRM_PSSWD=admin
JOIN_NODE_NAME=
DOMAIN_NAME=Domain
PCRS_DBTYPE_AZURE=0
CATALOGUE_SERVICE_SOLR_KEYSTORE_PASSWD=
DB_TYPE=Oracle
DIS_TRUSTSTORE_PASSWD=
KERBEROS_SECURITY_DOMAIN_NAME=
PASSWORD_COMPLEXITY=false
MRS_DB_SERVICENAME=
TAR_FILE_LOC=
ARCHIVE_FILE=
PCRS_SERVICE_NAME=
INFA_SERVICES_INSTALLED=false
DIS_KEYSTORE_PASSWD=
DPS_CLUSTER_TRUSTSTORE_FILE_PASSWD=
YARN_SERVICE_PRINCIPAL=
CMS_HTTPS_PORT=
CATALOGUE_SERVICE_TLS_HTTPS_PORT=
EDP_MRS_DB_TYPE=
KDC_TYPE=mit-kdc/active-directory
MRS_DB_ADDRESS=
MRS_DB_CUSTOM_STRING_SELECTION=0
CMS_PROTOCOL_TYPE=http
PWH_DB_CUSTOM_STRING=
IHS_ADMINISTRATOR_SERVER_HOST=
CATALOGUE_SERVICE_CUSTOM_SELECTION=false
MONITORING_DB_PASSWD=
DPS_DIS_SERVICE_NAME=Data_Integration_Service_EDP
JOIN_DOMAIN=0
TLS_CUSTOM_SELECTION=false
HIVE_LOCALSTORAGE_FORMAT=DefaultFormat
PORT_RANGE=6005-6115
ENABLE_DPS_SERVICE=true
IMAGE_NAME=informaticaltd/dei:1040-rc336.update
CATALOGUE_SERVICE_NAME_ASSOCIATED=catalog_service
TUNE_APPLICATION_SERVICES=true
EDP_MRS_DB_PASSWD=
DPS_MRS_SERVICE_NAME=Model_Repository_Service_EDP
ENABLE_BIGDATA_JOB_RECOVERY=
DB_CUSTOM_STRING_SELECTION=0
EDP_NODE_NAME=Node01
MONITORING_DB2_TABLESPACE=
CREATE_MONITORING_STATS=0
CLUSTER_NAME_CCO=
CREATE_SERVICES=0
IHS_REALM=
EDP_KEYSTORE_DIR=
MONITORING_SERVICE_NAME=
EIC_SERVICES_INSTALLED=false
DPS_SERVICE_NAME=Interactive_Data_Preparation_Service
KDC_DOMAIN_NAME=PLATFORMKRB.COM
ZOOKEEPER_URI=
JOIN_DOMAIN_PORT=
CLUSTER_USERID=
DPS_DB_PSSWD=
EDP_MRS_DB_SSL_ENABLED=
LDAP_URL=
YARN_RESOURCE_MANAGER_URI=
EDP_DIS_HTTP_PORT=
MRS_DB_UNAME=admin
DPS_DB_SERVICENAME=
PCRS_DB_SERVICE_NAME=
HADOOP_KEYSTORE_ALIAS=
INSTALL_OPTION=2
MONITORING_DB_TYPE=Oracle
CREATE_CCO_CONNECTION=0
IS_CLUSTER_SECURE=false
DIS_KEYSTORE_DIR=
HDFS_LOCATION=/datalake/DPS
```

```
CATALOG_ENABLE_EMAIL_SERVICE=false
JOIN_HOST_NAME=
HADOOP_SERVICE_PORT=8160
PWH_DB2_TABLESPACE=
EXTERNAL_CLUSTER_TRUSTSTORE_FILE_LOC=
RUN_CLUSTER_PREVALIDATION=0
CLUSTER_HOST=
HADOOP_SERVICE_NAME=hadoop_service
PWH_DB_UNAME=CPI_PWH
productType=PC
CMS_DB_PASSWD=CPI_CMS
DOMAINMACHINE_PASSWORDLESS_SSH_ENABLED=0
PASS_PHRASE_PASSWD=key
SLIDER_MAX_PORT=
TRUSTED_CONNECTION=
CMS_DB2_TABLESPACE=
ASSOCIATE_PROFILE_CONNECTION=0
CMS_DB_ADDRESS=insrh74dsg015.informatica.com:1521
PCRS_DB_USER=
DPS_DB_CUSTOM_STRING=
CREATE_EDP_SERVICES=0
ADVANCE_JDBC_PARAM=
CLUSTER_HADOOP_DISTRIBUTION_URL_USER=
HISTORY_SERVER_HTTP_URI=
EDP_KEYSTORE_PSSWD=
HADOOP_LOG_DIR=
MONITORING_DB_CUSTOM_STRING_SELECTION=0
DOMAIN_KEYSTORE_ALIAS=
TOMCAT_PORT=
EXTERNAL_CLUSTER_TRUSTSTORE_FILE_PASSWD=
GATEWAY_USERNAME=root
SLIDER_MIN_PORT=
```

Author

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